

Math 403 Homework #1, Spring 2025

Instructor: Ezra Miller

Solutions by: ...your name...

Collaborators: ...list those with whom you worked on this assignment...
(1 point for each of up to 3 collaborators who also list you)

/3

Due: 11:59pm Saturday 25 January 2025

READING ASSIGNMENTS (item numbers are lecture numbers as in PDF lecture notes)

/57

1. for Thu. 9 January
 - [Hefferon, p.153–154] field axioms
 - [Cornell, “Fields” (about 1/3 of the way through 4330-week1.pdf)]: more on fields
 - [Climenhaga, §7.2, §9.1] isomorphism, rank-nullity
2. for Tue. 14 January
 - [Lax, Chapter 1] quotients
 - [Climenhaga, §5.1] quotients
 - [Cornell, “Quotient Spaces” (4330-week5.pdf)] universal properties
 - [Cornell, “Exact Sequences” (about halfway through 4330-week5.pdf)]
3. for Thu. 16 January
 - [Lax, Chapter 2] duality
 - [Climenhaga, §5.1, §8.2] duality, transpose
 - [Treil, §5.1–5.5] Hermitian inner products, adjoints; §5.2–5.4 should be mostly review
4. for Tue. 21 January
 - [Lax, Appendix 15] Jordan canonical form (very short proof)
 - [Hefferon, §5.IV.1] characteristic polynomial and minimal polynomial
 - [Treil, §9.3 and §9.5] generalized eigenspaces and Jordan canonical form
5. for Thu. 23 January
 - [Lax, Chapter 14, p.214–221] norms, equivalence, continuity, local compactness
 - [Stewart–Sun, §II.1] norms, equivalence, Hahn–Banach theorem
6. for Tue. 28 January
 - [Lax, Chap.12, p.187–190] Convex sets
7. for Thu. 30 January
 - [Serge Lang, *Linear Algebra*, Chapter XII, §1–§2] separating hyperplanes
 - [Serge Lang, *Linear Algebra*, Chapter XII, §3–§4] support hyperplane, extreme point

EXERCISES

1. (Freshman's Dream): The *characteristic* of a field F is the smallest positive integer p such that the sum $1 + \cdots + 1$ of p multiplicative identities is 0 in F . Prove that p is prime if it is finite. If F has characteristic p , show that $(a + b)^p = a^p + b^p$ for $a, b \in F$. /3
2. Fix a vector space V and a subspace $W \subseteq V$ over a field F . Let $\pi : V \rightarrow V/W$ be the *projection homomorphism* given by $\pi(v) = v + W$. Write X for the set of all subspaces of V that contain W , and write Y for the set of all subspaces of V/W . Prove that π induces a bijection between these two sets, with /3

$$\begin{aligned} X &\rightarrow Y \\ L &\mapsto \pi(L) = \{\pi(v) \mid v \in L\} \end{aligned}$$

and

$$\begin{aligned} Y &\rightarrow X \\ M &\mapsto \pi^{-1}(M) = \{v \in V \mid \pi(v) \in M\}. \end{aligned}$$

3. Show that giving an exact sequence $\cdots \rightarrow V_{i-1} \rightarrow V_i \rightarrow V_{i+1} \rightarrow \cdots$ is the same as giving a collection of short exact sequences $0 \rightarrow K_i \rightarrow V_i \rightarrow K_{i+1} \rightarrow 0$, one for each i . (The long exact sequence is said to be constructed by *splicing* the short exact sequences together.) /3
4. Rank-nullity theorem for exact sequences: Given an exact sequence /3

$$0 \rightarrow V_0 \rightarrow V_1 \rightarrow \cdots \rightarrow V_r \rightarrow 0,$$

prove that $\sum_{i=0}^r (-1)^i \dim V_i = 0$.

5. Rank-nullity for arbitrary complexes: Given a complex $0 \rightarrow V_0 \rightarrow V_1 \rightarrow \cdots \rightarrow V_r \rightarrow 0$, write $B_i = \text{im}(V_{i-1} \rightarrow V_i)$ and $Z_i = \ker(V_i \rightarrow V_{i+1})$. Prove that /3

$$\sum_{i=0}^r (-1)^i \dim V_i = \sum_i (-1)^i \dim H_i,$$

where $H_i = Z_i/B_i$ is the i^{th} homology of the complex. [In Exercise 4, $B_i = Z_i = K_i$, so $H_i = Z_i/B_i = 0$ for all i .]

6. Two elements u and v in a vector space V are *congruent modulo* a subspace $W \subseteq V$, written $u \equiv v \pmod{W}$, if $u + W = v + W$. Show that congruence modulo W is an *equivalence relation* on V , meaning that it is /3
 - reflexive: $v \equiv v \pmod{W}$ for all $v \in V$;
 - symmetric: if $u \equiv v \pmod{W}$, then $v \equiv u \pmod{W}$; and
 - transitive: if $u \equiv v \pmod{W}$ and $v \equiv x \pmod{W}$ then $u \equiv x \pmod{W}$.

7. Give an example of three subspaces Y_1 , Y_2 , and Y_3 in $\mathbb{R}^2$ such that $Y_1 + Y_2 + Y_3 = \mathbb{R}^2$ and $Y_i \cap Y_j = \{\mathbf{0}\}$ for all $i \neq j$, but $\mathbb{R}^2$ is not the direct sum of Y_1 , Y_2 , and Y_3 . /3
8. Prove that if V is a vector space and $W \subseteq V$ is a subspace, then W has a *complement*: a subspace $U \subseteq V$ such that $V = W \oplus U$. Hint: V/W has a basis; lift it back to V . /3
9. For vectors $\mathbf{x} = (1, 2i, 1 + i)$ and $\mathbf{y} = (i, 2 - i, 3)$, compute
 - (a) $\langle \mathbf{x}, \mathbf{y} \rangle$, $\|\mathbf{x}\|^2$, $\|\mathbf{y}\|^2$, and $\|\mathbf{y}\|$; /3
 - (b) $\langle 3\mathbf{x}, 2i\mathbf{y} \rangle$ and $\langle 2\mathbf{x}, i\mathbf{x} + 2\mathbf{y} \rangle$; /3
 - (c) $\|\mathbf{x} + 2\mathbf{y}\|$. [Use parts (a) and (b) for this.] /3
10. Prove that for vectors in an inner product space, $\|\mathbf{x} \pm \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 \pm 2\operatorname{Re}\langle \mathbf{x}, \mathbf{y} \rangle$. /3
11. For any $m \times n$ complex matrix A , prove that $\ker(A^*A) = \ker(A)$. /3
12. Prove that if P is self-adjoint (that is, $P^* = P$) and idempotent (that is, $P^2 = P$) then P is the matrix for an orthogonal projection. /3
13. If V is a vector space over $\mathbb{C}$ of dimension n , then it is also a vector space over $\mathbb{R}$ of dimension $2n$. (If this isn't clear to you, then write down a proof.) Given a Hermitian inner product $\langle \mathbf{x}, \mathbf{y} \rangle$ on V as a complex vector space, show that the real part $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbb{R}} = \operatorname{Re}(\langle \mathbf{x}, \mathbf{y} \rangle)$ is an inner product on V as a real vector space. /3
14. Find all possible Jordan forms of linear transformations with characteristic polynomial $(t - 1)^2(t + 2)^2$. /3
15. Find all possible Jordan forms of linear transformations with characteristic polynomial $(t - 2)^3(t + 1)$ and minimal polynomial $(t - 2)^2(t + 1)$. /3
16. How many similarity classes are there for 3×3 matrices whose only eigenvalues are -3 and 4 ? /3
17. Prove or disprove: two $n \times n$ matrices are similar if and only if they have the same characteristic polynomial and minimal polynomial. /3